

Original scenario

Verdant Grocers is a regional chain. Store managers currently reorder from a report that is refreshed overnight, so a product that sells out at 10 AM is not visible as out-of-stock to the replenishment team until the next morning.

What is known:

- 140 stores across three states.
- Each store has point-of-sale terminals that already emit a transaction event per basket. The events reach an on-premises message broker today.
- Roughly 900,000 basket events per day across the chain, with a pronounced weekday-evening peak.
- The merchandising team wants store-level stock positions visible "within a few minutes" of a sale, and has said that a figure closer to one minute would change how they staff replenishment.
- There is an existing overnight batch warehouse that finance depends on. It cannot be switched off, and finance's monthly close must not regress.
- The company has two full-time data engineers. There is no platform team.
- Leadership has said it does not want to be "locked into one vendor", but has not defined what that means in practice.

Phase 2 additions from PR comment 5818993306

- Zero basket-event loss is required.
- Finance reconciles store sales against this event stream, and a missing basket is a missing sale on the books.
- A gap must be recoverable, not merely detected.
- A cloud run-rate ceiling of USD 8,000 per month for the new platform, excluding the existing overnight warehouse, which is budgeted separately.
- PCI DSS applies to the POS transaction stream.
- Cardholder data must never land in the analytics platform; it has to be removed or tokenised before the events leave the store network.

Phase 3 correction from PR comment 5820970476

- Each POS terminal keeps a local journal of every basket it rings, retained for 72 hours.
- If the streaming path drops events during an outage, the journal is re-sent in the overnight batch and finance reconciles from that.
- The business therefore tolerates loss of up to roughly 15 minutes of basket events from the near-real-time inventory view during an outage, provided the loss is visible to merchandising and is corrected by the next morning.